

Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Solving a System Using a Graph

Toolbox

Solving a System Using a Graph

Solving a system means finding the values of the variables. One way to solve a system is by graphing the equations.

Example

Hide

Solve:
$$\begin{cases} y = -x + 3 \\ y = \frac{1}{2}x - 3 \end{cases}$$

Step 1

Graph both equations on the same grid.
Remember, you can always graph a line by making a table of values and then plotting points.

$y = -x + 3$

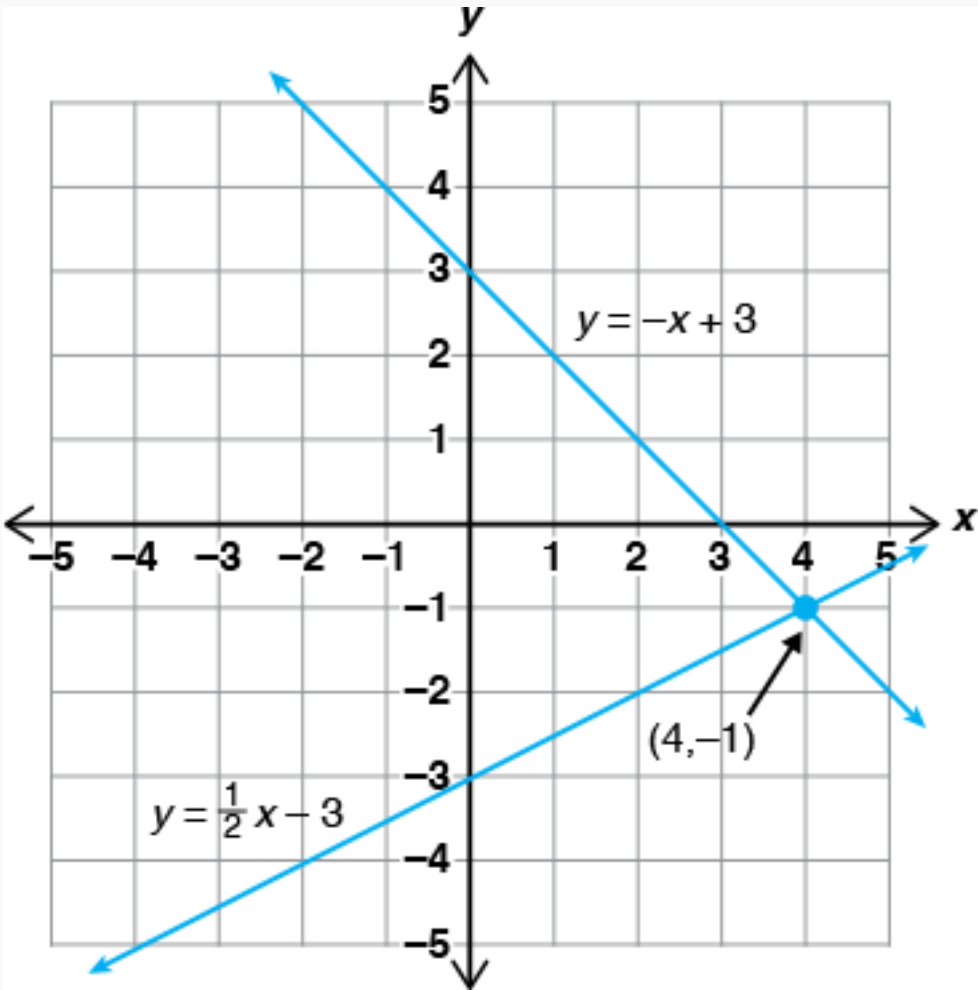
<i>x</i>	<i>y</i>
−1	4
0	3
1	2

$y = \frac{1}{2}x - 3$

<i>x</i>	<i>y</i>
−2	−4
0	−3
2	−2

Step 2

The lines intersect at point (4, −1). The *x*- and *y*-coordinates of this point are the solution to the system: $x = 4$ and $y = -1$.



Step 3

Check your answer. Substitute the values into the system. Make sure both equations are true.

$y = -x + 3$
 $-1 = -(4) + 3$
 $-1 = -1$ True

$y = \frac{1}{2}x - 3$
 $-1 = \frac{1}{2}(4) - 3$
 $-1 = 2 - 3$
 $-1 = -1$ True

Key Term

system of equations

Report Content Errors

Rate Your Confidence

High

Medium

Low

Continue >>

Category

Lesson 5: Graphing Systems of Equations and Inequalities

Related Questions:

[Practice questions taken related to this topic](#)